

→ Coordinate frames

> a frame defines:

- its origin
- positive x, y, z directions
- its measurement units

transformation:

$$P_c = R \cdot P_m + t$$

1. If the marker rotates around its center, which mainly changes: R or t ?
2. Why is t alone insufficient to describe the complete marker pose?
3. Does rotating the marker change the camera frame itself?

1. no moving along the axes only rotation: R
2. marker may be tilted so we need R too
3. cam. frame does not change when only the marker rotates
↳ camera frame stays attached to the camera
↳ marker frame rotates w/ the marker

Think of a stationary map and a rotating car: the car's direction relative to the map changes, but the map itself doesn't rotate.

In our experiments, we've already used three kinds of frames:

- Chessboard/world frame: origin and axes attached to the chessboard.
- Marker frame: origin at the ArUco center; axes rotate with the marker.
- Camera frame: origin at the camera; $+X$ right, $+Y$ down, $+Z$ forward.

"World frame" doesn't mean the universe's true coordinate system. It is simply a convenient fixed reference we choose—such as a chessboard corner, table corner, or robot base.

1. A stationary marker is centered 400 mm ahead. Is its t_{vec} zero?
2. When would $t_{vec} = [0, 0, 0]$?
3. If the marker moves 100 mm farther forward, which component changes, and by how much?

1. no, z is 400mm
2. when the camera's origin == marker's origin
3. z increases by 100mm

2. 2 angles + 100 mm

$$\rightarrow R \cdot \overset{\text{marker}}{[1, 0, 0]} = \overset{\text{cam.}}{[0, 1, 0]}$$

$\rightarrow R$: explains what the marker's axis directions mean when described using the camera's axes.

$\hookrightarrow$ changes direction

$\hookrightarrow$ preserves: length, angles, shape

1. If marker and camera axes are perfectly aligned, what is R ?
2. If R maps marker $+X$ to camera $+Y$, where does marker-right point in the camera frame?
3. Can a valid rotation matrix turn a 30 mm vector into a 60 mm vector?

1. identity

2. downward

3. no length stays same

Columns of describes the rotated axes:

$$R = [\text{marker } +X \mid \text{marker } +Y \mid \text{marker } +Z]$$

$$R = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\hookrightarrow$ marker $+X \rightarrow$ camera $[0, 1, 0] \rightarrow$ cam-down

marker $+Y \rightarrow$ camera $[-1, 0, 0] \rightarrow$ cam-left

marker $+Z \rightarrow$ camera $[0, 0, 1] \rightarrow$ cam-forward

1. Since marker $+X$ maps to camera $+Y$, what is:

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R · [30, 0, 0]

[10, 14, 300]

2. Now add **t** to your first answer. What is **P_c**?

3. In plain language, where is this point relative to the camera?

1. [0, 30, 0]

2. [10, 44, 300]

3. 10 right, 44 down, 300 forward

Homogeneous Transformation

$$T_{\text{camera} \leftarrow \text{marker}}: \begin{bmatrix} R & \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[x, y, z] \rightarrow [x, y, z, 1]$$

3x3

4x4

distinguishes

point: [x, y, z, 1]

direction: [x, y, z, 0]

> take coords. from the marker frame
and express in the camera frame

$$P_c = T_{\text{camera} \leftarrow \text{marker}} \cdot P_m$$

transform
marker
origin

$$[0, 0, 0] \rightarrow [t_x, t_y, t_z, 1]$$

for a point: $T \cdot [p, 1] \rightarrow R \cdot p + t$ → the translation is applied

for a direction: $T \cdot [d, 0] \rightarrow R \cdot d$ → the translation mult. by 0
not applied

```
R = identity
t = [10,14,300]
```

Answer:

1. What happens to the point $[30,0,0,1]$?
2. What happens to the direction $[30,0,0,0]$?
3. Which homogeneous value receives translation: $w=1$ or $w=0$?

1. $[30,0,0,1] \rightarrow [40,14,300,1]$

2. $[30,0,0,0] \rightarrow [30,0,0,0]$

3. $w=1$

inverse

going backwards; $\rightarrow$ useful when
marker is fixed, camera is not fixed
 $T_{\text{marker} \leftarrow \text{camera}} = \text{inverse}(T_{\text{camera} \leftarrow \text{marker}})$ "moving"
from PnP

$$R \rightarrow R^T$$

$$t \rightarrow -R^T \cdot t$$

$$T^{-1} = \begin{bmatrix} R^T & | & -R^T t \\ \hline 0 & 0 & 0 & | & 1 \end{bmatrix}$$

1. If the forward transform converts marker coordinates into camera coordinates, what does its inverse convert?
2. What operation gives the inverse of a rotation matrix?
3. If R is identity, what does the inverse translation simplify to?

1. cam. to marker

2. R^T

3. it becomes $-t'$

> now, to connect multiple frames

$T_{\text{camera} \leftarrow \text{marker}}$
comes from ArUco PnP

$T_{\text{robot} \leftarrow \text{camera}}$
from mounting the cam. onto a robot

$\Rightarrow T_{\text{robot} \leftarrow \text{marker}}$

$= T_{\text{robot} \leftarrow \text{camera}} \cdot T_{\text{camera} \leftarrow \text{marker}}$

$= \text{marker} \rightarrow \text{camera} \rightarrow \text{robot}$
connects the
transformations

$P_{\text{robot}} = T_{\text{robot} \leftarrow \text{camera}} \cdot T_{\text{camera} \leftarrow \text{marker}} \cdot P_{\text{marker}}$

Where is the marker relative to the camera? $\rightarrow$ use PnP directly

Where is the camera relative to the marker? $\rightarrow$ invert PnP

Where is the marker relative to the robot?

$\rightarrow T_{\text{robot} \leftarrow \text{camera}} \cdot T_{\text{camera} \leftarrow \text{marker}}$

extend
the
chain

marker $\rightarrow$ camera $\rightarrow$ robot $\rightarrow$ world

$$T_{\text{world} \leftarrow \text{marker}} = T_{\text{world} \leftarrow \text{robot}} * T_{\text{robot} \leftarrow \text{camera}} * T_{\text{camera} \leftarrow \text{marker}}$$

1. PnP converts marker coordinates into camera coordinates.
2. Camera mounting calibration converts camera coordinates into robot coordinates.
3. Robot localization converts robot coordinates into world coordinates.

suppose all axes are aligned, meaning every 'R' is identity:

marker $\rightarrow$ camera: $[0, 0, 300] \text{ mm}$

camera $\rightarrow$ robot: $[50, 0, 0] \text{ mm}$

robot $\rightarrow$ world: $[1000, 0, 0] \text{ mm}$

marker $\rightarrow$ world: $[1050, 0, 300] \text{ mm}$

composition does this
when rotation exists:

$$T_{\text{robot} \leftarrow \text{camera}} = [R_{rc} | t_{rc}]$$

$$T_{\text{camera} \leftarrow \text{marker}} = [R_{cm} | t_{cm}]$$

> combined translation:

$$t_{rm} = R_{rc} \cdot t_{cm} + t_{rc}$$

> marker's cam-relative
translation must first
be rotated into
robot-axis
dir.

Concrete example:

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$t_{cm} = [0, 0, 300]$

The marker is 300 mm along camera +Z.

Suppose the camera is mounted sideways, so:

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Camera +Z $\rightarrow$ Robot +X

Then:

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$R_{rc} \cdot [0, 0, 300] = [300, 0, 0]$

t_{cm}

If the camera origin is also located at

Summary

```
rvec
→ rotation matrix R
→ combine R and t into 4x4 Tcamera←marker
→ invert it into Tmarker←camera
→ multiply T-1T
→ recover the identity matrix
```

- PnP provides `rvec, tvec`.
- Rodrigues converts `rvec` into `R`.
- We construct the homogeneous transformation.
- We calculate its inverse.
- We verify that applying a transform and its inverse brings us back to the starting frame.

Conclusion

$$P_c = R \cdot P_m + t$$

- A 4x4 homogeneous matrix combines `R` and `t`.
- Points use `w=1` and receive translation.
- Directions use `w=0` and only rotate.
- PnP gave `Tcamera←marker`.
- Its inverse gave `Tmarker←camera`.
- Inverse translation is `-RTt`, not simply `-t`.
- Transformations compose right to left.
- Transform followed by its inverse returned identity.
- A local marker point was successfully converted into camera coordinates.

If the camera origin is also located at:

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`trc = [50,0,0]`

the marker's robot coordinates become:

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$$t_{rm} = R_{rc} \cdot t_{cm} + t_{rc}$$
$$t_{rm} = [300,0,0] + [50,0,0]$$
$$= [350,0,0]$$